

AP Statistics FRQ Practice Exam 03

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AP Statistics FRQ Practice Exam 3

APSTAT-FRQ3-001

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: Unit 3, Inference for Categorical Data: Proportions

FRQ Type: Formulate Questions and Collect Data

Prompt:

A county water-conservation office wants to estimate the proportion of all rain barrels at county residences that are properly maintained. The county contains 48 neighborhoods and has a registry of 1,920 rain barrels. Some residential rain barrels are not registered.

The office randomly selects 8 of the 48 neighborhoods and inspects every registered rain barrel in each selected neighborhood. The results are shown.

Selected neighborhood	A	B	C	D	E	F	G	H	Total
Registered barrels	28	33	41	35	26	47	39	37	286
Properly maintained	20	26	34	27	18	39	28	30	222

- (a) Identify the population of interest and the sample actually inspected.
- (b) Identify the sampling method. Explain how the random selection is carried out in this method.
- (c) Calculate the sample proportion of inspected barrels that are properly maintained. Identify the population parameter it is intended to estimate, and state the statistical question being investigated.
- (d) Explain how using the registry can create undercoverage bias. If unregistered barrels are less likely to be properly maintained than registered barrels, state the likely direction of the bias.
- (e) A staff member claims that this sample of 286 barrels must be more precise than a simple random sample of 286 registered barrels. Evaluate the claim, assuming maintenance practices tend to be similar for barrels within the same neighborhood.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Identifies the population of interest as all rain barrels at county residences, including registered and unregistered barrels.
- Point 2: Identifies the sample as the 286 registered rain barrels inspected in the 8 randomly selected neighborhoods.

(b) 2 points

- Point 1: Identifies the method as a one-stage cluster sample, with neighborhoods serving as clusters.
- Point 2: Explains that 8 neighborhoods are randomly selected from the 48 and that every registered barrel in each chosen neighborhood is inspected.

(c) 2 points

- Point 1: Calculates $\hat{p} = 222/286 \approx 0.776$, or 77.6%, and identifies p as the true proportion of all county residential rain barrels that are properly maintained.
- Point 2: States an appropriate statistical question, such as, “What proportion of all rain barrels at county residences are properly maintained?”

(d) 2 points

- Point 1: Explains that barrels missing from the registry have no chance to be selected, so the sampling frame undercovers the target population.
- Point 2: Concludes that $\hat{p}$ would tend to overestimate the population proportion if unregistered barrels are less likely to be properly maintained.

(e) 2 points

- Point 1: Rejects the assertion that the cluster sample must be more precise solely because it contains 286 barrels; sample size alone does not determine precision when observations are clustered.
- Point 2: Explains that similar maintenance practices within neighborhoods provide less independent information than a geographically dispersed SRS, so the cluster estimate may have greater sampling variability; an SRS of registered barrels could be more precise.

Model Guidance:

The target includes every residential rain barrel in the county, but the inspected sample contains only registered barrels in eight selected neighborhoods. Selecting whole neighborhoods and inspecting all registered barrels within them is cluster sampling. The observed proportion is $222/286 \approx 0.776$, intended to estimate the countywide maintenance proportion.

The registry is not a complete frame for the stated population. If unregistered barrels are systematically less well maintained, excluding them makes the estimate too high. Clustering creates a separate precision issue: barrels in one neighborhood may share weather, outreach programs, or local maintenance habits. Many similar observations from a few neighborhoods may provide less information than the same number of barrels spread across the county. A complete address-based frame or a method for locating unregistered barrels would address undercoverage; sampling more neighborhoods would generally improve geographic representation.

Expected Points:

10

APSTAT-FRQ3-002

Section: FRQ

Type: frq

Primary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Secondary Unit: None

FRQ Type: Analyze Data, One- or Two-Variable Data and Probability

Prompt:

On a clear night, a remote observatory plans 18 six-minute telescope exposures. Long-term records indicate that an exposure has probability 0.12 of being spoiled by a satellite trail. For this question, first assume that the outcomes of different exposures are independent and that the probability remains 0.12. Let X be the number of spoiled exposures among the 18.

- (a) Explain why X can be modeled by a binomial distribution, and state its parameters.
- (b) Calculate $P(X = 2)$ and $P(X \leq 2)$.
- (c) Calculate the mean and standard deviation of X . Interpret both values in context.
- (d) The observatory director claims that it is more likely than not that at least 3 of the 18 exposures will be spoiled. Calculate the relevant probability and evaluate the claim.
- (e) On some nights, satellite trails arrive in bursts, making nearby exposures more likely to be spoiled together. Explain which binomial condition is violated and why the original binomial probabilities may then be unreliable.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Verifies a fixed number of 18 trials, two outcomes for each exposure, and a constant spoilage probability of 0.12.
- Point 2: Cites independence and states $X \sim \text{Binomial}(n = 18, p = 0.12)$.

(b) 2 points

- Point 1: Calculates $P(X = 2) = C(18, 2)(0.12)^2(0.88)^{16} \approx 0.2850$.
- Point 2: Calculates $P(X \leq 2) = \sum \text{from } k = 0 \text{ to } 2 \text{ of } C(18, k)(0.12)^k(0.88)^{18-k} \approx 0.6310$.

(c) 2 points

- Point 1: Calculates $\mu_X = np = 18(0.12) = 2.16$ exposures and $\sigma_X = \sqrt{np(1 - p)} = \sqrt{18(0.12)(0.88)} \approx 1.38$ exposures.
- Point 2: Interprets 2.16 as the long-run mean number spoiled per set of 18 exposures and 1.38 as the typical distance of X from that mean across many comparable clear nights.

(d) 2 points

- Point 1: Calculates $P(X \geq 3) = 1 - P(X \leq 2) = 1 - 0.6310 = 0.3690$.
- Point 2: Rejects the “more likely than not” claim because $0.3690 < 0.50$; under the model, three or more spoiled exposures occur less than half the time.

(e) 2 points

- Point 1: Identifies independence as the violated binomial condition because the status of one exposure changes the probability that a nearby exposure is spoiled.
- Point 2: Explains that clustering of spoiled exposures changes the distribution of X , typically producing more night-to-night variability than the independent binomial model, so probabilities such as $P(X \geq 3)$ need not equal the calculated value.

Model Guidance:

The setting has 18 fixed trials, each classified as spoiled or not spoiled, with constant $p = 0.12$ and assumed independence. Thus X has a $\text{Binomial}(18, 0.12)$ distribution. Substituting into the binomial formula gives $P(X = 2) \approx 0.2850$; adding the probabilities for 0, 1, and 2 gives about 0.6310.

Over many comparable nights, the number spoiled averages 2.16 per 18 exposures and typically differs from that average by about 1.38. The complement calculation gives $P(X \geq 3) \approx 0.3690$, which does not support a probability greater than one-half. If trails arrive in bursts, neighboring exposure outcomes are associated rather than independent. The binomial model then understates clustering and cannot be trusted without a model that accounts for that dependence.

Expected Points:

10

APSTAT-FRQ3-003

Section: FRQ

Type: frq

Primary Unit: Unit 4, Inference for Quantitative Data: Means

Secondary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

FRQ Type: Analyze Data, Statistical Inference (Confidence Interval or Significance Test)

Prompt:

A credit union states that the mean time required to resolve all card-fraud disputes closed during the past year was less than 10 days. An auditor takes a simple random sample of 25 from the 3,200 disputes closed during that year. The sample mean resolution time is $\bar{x} = 8.7$ days, and the sample standard deviation is $s = 3.2$ days. A histogram of the 25 resolution times is moderately right-skewed but has no apparent outliers.

At the $\alpha = 0.05$ significance level, test the credit union's claim.

- Define the population parameter and state the hypotheses.
- Identify the appropriate inference procedure and check its conditions.
- Calculate the standard error, test statistic, degrees of freedom, and p-value.
- State a conclusion in context using the $\alpha = 0.05$ significance level.
- Describe a Type I error in the context of this test.

Scoring Guide:

Total Points: 10

(a) 2 points

- Point 1: Defines μ as the true mean resolution time, in days, for all 3,200 card-fraud disputes closed by the credit union during the past year.
- Point 2: States $H_0: \mu = 10$ days and $H_a: \mu < 10$ days.

(b) 3 points

- Point 1: Selects a one-sample t test for a population mean because σ is unknown and the data are quantitative.
- Point 2: Cites the SRS and verifies the 10% condition because $25 \leq 0.10(3,200)$, supporting independence of observations.
- Point 3: Explains that a sample of 25 with only moderate skewness and no apparent outliers is adequate for a t procedure, which is reasonably robust in this setting.

(c) 3 points

- Point 1: Calculates $SE = s/\sqrt{n} = 3.2/\sqrt{25} = 0.64$ day.
- Point 2: Calculates $t = (8.7 - 10)/0.64 \approx -2.03$ with $df = 25 - 1 = 24$.
- Point 3: Calculates the left-tail p -value $P(T_{24} \leq -2.03) \approx 0.0267$.

(d) 1 point

- Point 1: Because $0.0267 < 0.05$, rejects H_0 and concludes that the sample provides convincing evidence that the true mean resolution time for disputes closed during the past year was less than 10 days.

(e) 1 point

- Point 1: Describes a Type I error as concluding that the true mean resolution time was less than 10 days when it was not less than 10 days.

Model Guidance:

The claim concerns one population mean and the population standard deviation is unknown, so a one-sample t test is appropriate. The random sample and 10% condition support independence. With $n = 25$, moderate right skew without outliers is not severe enough to prevent use of the t procedure.

The estimated standard error is $3.2/5 = 0.64$ day. The test statistic is $(8.7 - 10)/0.64 \approx -2.03$ with 24 degrees of freedom, producing a one-sided p -value of about 0.0267. If the true mean were 10 days, a sample mean at least this far below 10 would occur about 2.67% of the time under repeated random sampling. Since that probability is below 0.05, the data support the credit union's claim for the population and time period stated. A Type I error would mean accepting that claim even though the population mean was not below 10 days.

Expected Points:

10

APSTAT-FRQ3-004

Section: FRQ

Type: frq

Primary Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Secondary Unit: Unit 2, Probability, Random Variables, and Probability Distributions

FRQ Type: Interpret Results, Integrated Analysis and Justification

Prompt:

A coffee-roasting laboratory studies the percentage of initial mass lost during roasting. Eighty comparable test batches are randomly assigned, 40 to a Constant-Heat profile and 40 to a Pulse-Heat profile. Summary statistics are shown.

Statistic	Constant-Heat	Pulse-Heat
Minimum	10.8%	10.9%
First quartile, Q1	11.6%	11.8%
Median	12.1%	12.2%
Third quartile, Q3	12.7%	12.5%
Largest nonoutlying value	13.4%	13.1%
Maximum	15.4%	13.1%
Mean	12.3%	12.2%
Standard deviation	0.90%	0.55%

(a) Calculate the IQR and the $1.5 \times \text{IQR}$ outlier fences for each profile. Determine whether the maximum value for either profile is a potential outlier.

(b) On paper, sketch side-by-side modified boxplots using the table. Use a common labeled scale and show any potential outlier separately. Then compare the distributions with respect to center, spread, and unusual features.

(c) A technician claims that Pulse-Heat produces more consistent mass loss without materially changing the typical mass loss. Evaluate this claim using at least two appropriate statistics from the table.

(d) For future Pulse-Heat batches made under similar conditions, suppose mass-loss percentage Y can be modeled as Normal with $\mu = 12.2\%$ and $\sigma = 0.55\%$. Calculate $P(11.4\% \leq Y \leq 13.0\%)$ and interpret the result.

(e) A production manager claims that the Pulse-Heat profile causes at least 85% of comparable batches to meet the 11.4% to 13.0% target. Use the experimental design and your result from part (d) to evaluate the claim. Include one limitation or condition in your reasoning.

Scoring Guide:

Total Points: 10

(a) 2 points

• Point 1: For Constant-Heat, calculates $\text{IQR} = 12.7 - 11.6 = 1.1$, fences 9.95 and 14.35, and identifies 15.4% as a potential high outlier.

• Point 2: For Pulse-Heat, calculates $\text{IQR} = 12.5 - 11.8 = 0.7$, fences 10.75 and 13.55, and concludes that the 10.9% minimum and 13.1% maximum are not potential outliers.

(b) 2 points

- Point 1: Draws modified boxplots on a common percentage scale with correct quartiles and medians, whiskers to 10.8% and 13.4% for Constant-Heat plus a separate point at 15.4%, and whiskers to 10.9% and 13.1% for Pulse-Heat.
- Point 2: Describes similar centers, substantially smaller spread for Pulse-Heat, and the high potential outlier and longer upper reach for Constant-Heat.

(c) 2 points

- Point 1: Supports greater consistency for Pulse-Heat using its smaller IQR, 0.7% versus 1.1%, and/or its smaller standard deviation, 0.55% versus 0.90%.
- Point 2: Supports little change in typical mass loss using the close medians, 12.2% versus 12.1%, and/or the close means, 12.2% versus 12.3%.

(d) 2 points

- Point 1: Standardizes the bounds as $z = (11.4 - 12.2)/0.55 \approx -1.45$ and $z = (13.0 - 12.2)/0.55 \approx 1.45$, then calculates $P(-1.45 \leq Z \leq 1.45) \approx 0.854$.
- Point 2: Interprets the result: under the normal model, about 85.4% of comparable future Pulse-Heat batches are expected to have mass loss from 11.4% through 13.0%.

(e) 2 points

- Point 1: Explains that random assignment supports a causal comparison of the two heat profiles for batches like those in the experiment, and that the model estimate of 85.4% is consistent with the “at least 85%” threshold.
- Point 2: Qualifies the claim because 85.4% is only slightly above 85% and is based on an assumed normal model with estimated parameters; without a confidence analysis it does not establish the population percentage above 85%, and generalization requires future batches and operating conditions to resemble those studied.

Model Guidance:

For Constant-Heat, the IQR is 1.1%, so the upper fence is 14.35%; the 15.4% maximum is plotted separately in a modified boxplot. Pulse-Heat has IQR 0.7% and no flagged values. The side-by-side display should therefore show nearly equal medians but a shorter Pulse-Heat box and overall range, while Constant-Heat has an isolated high value. Both IQR and standard deviation support greater consistency for Pulse-Heat, and the means and medians support little change in center.

Under the stated normal model, the target endpoints are approximately -1.45 and 1.45 standard deviations from 12.2%, giving a central probability of about 0.854. Random assignment permits the observed distributional differences to be attributed to the heat profiles for comparable experimental batches. However, a fitted probability of 85.4% is not the same as statistically demonstrating that the true success rate exceeds 85%. The manager's claim is compatible with the model, but it depends on the normal assumption, parameter accuracy, and future conditions matching the experiment.

Expected Points:

10

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-FRQ3

- APSTAT-FRQ3-001: Unit 1 primary; Unit 3 secondary; one-stage cluster sample of every registered rain barrel in 8 randomly selected neighborhoods from 48, with 286 inspected and 222 properly maintained; neighborhood-by-count table; identify the population of all residential barrels and the inspected sample, calculate $\hat{p} = 222/286$, formulate the county maintenance-proportion question, diagnose registry undercoverage and its conditional direction, and compare cluster precision with an

SRS when within-neighborhood practices are similar; targets confusing cluster size with precision and overlooking unregistered units.

- APSTAT-FRQ3-002: Unit 2 primary; no secondary unit; Binomial(18, 0.12) model for telescope exposures spoiled by satellite trails on a clear night; verify binomial conditions, calculate $P(X = 2) \approx 0.2850$ and $P(X \leq 2) \approx 0.6310$, calculate and interpret $\mu_X = 2.16$ and $\sigma_X \approx 1.38$, reject a more-likely-than-not claim using $P(X \geq 3) \approx 0.3690$, and explain how burst arrivals violate independence and increase clustering; targets complement errors and mechanical use of a binomial model under dependence.
- APSTAT-FRQ3-003: Unit 4 primary; Unit 1 secondary; left-tailed one-sample t test for the mean card-fraud dispute resolution time among 3,200 credit-union cases closed in one year, using an SRS of 25 with $\bar{x} = 8.7$ days, $s = 3.2$ days, moderate right skew and no outliers; state $H_0: \mu = 10$ and $H_a: \mu < 10$, check SRS/10%/shape conditions, calculate $SE = 0.64$, $t \approx -2.03$, $df = 24$, $p \approx 0.0267$, reject at $\alpha = 0.05$, and describe a Type I error; targets using z with unknown σ and misreading a one-sided p-value.
- APSTAT-FRQ3-004: Unit 1 primary; Unit 2 secondary; completely randomized comparison of mass-loss percentages for 40 Constant-Heat and 40 Pulse-Heat coffee-roasting batches; summary table with quartiles, means, SDs, largest nonoutlying values, and a 15.4% Constant-Heat outlier; calculate IQR fences, draw comparative modified boxplots, justify greater Pulse-Heat consistency with similar center, use a Normal(12.2, 0.55) model to obtain $P(11.4 \leq Y \leq 13.0) \approx 0.854$, and evaluate a causal at-least-85% target claim with model and scope limitations; targets equating similar centers with similar distributions and treating a fitted probability just above a threshold as proof.